Ignacio Manzur

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@nachitomanzur

About

Second-year PhD at the EPFL COMPSEC Lab, advised by Alessandro Chiesa.
Previously worked as a Cryptography Researcher at Nethermind for around two years.
MASt in Pure Mathematics (Part III) from the University of Cambridge.
Broadly interested in coding theory, the interaction between complexity theory and probabilistic proof/argument systems, and quantum computing.

Publications

2026 All Polynomial Generators Preserve Distance with Mutual Correlated Agreement
with S. Bordage, A. Chiesa and Z. Guan, accepted at CCC 2026

2024 FLI: Folding lookup instances
with A. Garreta, accepted at Asiacrypt 2024

Preprints

2026 Mutual Correlated Agreement and List-Decoding: Improvements and Separations

2024 Mova: Nova folding without committing to error terms
with N. Dimitriou, A. Garreta, I.Vlasov

2024 On amortization techniques for FRI-based SNARKs
with A. Garreta, H. Hovhanissyan, A. Jivanyan, I. Villalobos, M. Zajac

Talks

August 2026 CCC, Lisbon
All Polynomial Generators Preserve Distance with Mutual Correlated Agreement

June 2026 ZKProof, Rome
All Polynomial Generators Preserve Distance with Mutual Correlated Agreement

September 2024 TUM Blockchain Conference, Munich
Mova: Nova folding without committing to error terms

December 2023 Nethermind Internal Research Seminar
Presentation of the Basefold paper

April 2023 Ethereum Zürich Conference, Zürich
Simulation Extractability of zk-SNARKs

2021 Part III Seminar, Cambridge
Introduction to the Étale fundamental group

2021 Warwick–Imperial Mathematics Conference (WIMP)
The Riemann-Roch theorem, and Weil's Conjectures for curves

Teaching

Spring 2026 — Teaching Assistant, Theory of Computation
EPFL

Spring 2026 — Project supervision
Supervised Paul Quesnot's Bachelor's project, *WHIR accumulation through lattice-based folding*.
Supervised Loris Tran's Master's project, *A WHIR-backed succinct root proof for WARP accumulation*.

Service

2026 — External reviewer, Asiacrypt

2025 — External reviewer, IEEE Transactions on Information Theory

Conferences

August 2026 — CCC, Lisbon

June 2026 — ZKProof, Rome

January 2025 — Zinal Winter School, Zinal
How to delegate and verify quantum computations (Thomas Vidick)

December 2024 — TCC, Milan

September 2024 — TUM Blockchain Conference, Munich

August 2024 — SBC, Columbia

May 2024 — zkProof, Berlin

April 2024 — zkSummit11, Athens

September 2023 — zkSummit10, London

August 2023 — SBC, Stanford

July 2023 — zkWeek, Barcelona

July 2023 — DeFi Security Summit, MEVDay, Paris

May 2023 — EDCON, Podgorica
Participated as a judge during the SuperDemo, an event where more than 40 innovative projects in zk/privacy and decentralized applications showcase their ideas and compete for a prize pool.

April 2023 — Ethereum Zürich Conference, Zürich

February 2023 — Bar-Ilan Winter school on cryptography, Tel-Aviv
Blockchain technologies.
The winter school explained the cryptographic and economic concepts involved in blockchain from first principles, and then moved on to recent advances in blockchain technology.

Work experience

November 2024 – today — Doctoral Assistant
EPFL COMPSEC Lab, Lausanne

July 2022 – November 2024 Blockchain and Cryptography Researcher

Nethermind Research, London

- Investigator in a project aimed at extending the construction of Data Availability Schemes based on the FRI IOPP to work up to the Johnson bound.
- Investigator in a project aimed at developing Mova, the state-of-the-art folding scheme for R1CS instances.
- Investigator in a project aimed at developing FLI, an efficient folding scheme for lookup arguments.
- Investigator in a project aimed at developing efficient proof generation methods for STARKs.
- Investigator in the research effort aimed at developing a specification for v.2 of Obol's Distributed Validator middleware (Charon).
- Lead investigator in 0xPARC grant *Simulation Extractability of zk-STARKs*
- Investigator in a multi-phased project aimed at helping LIDO decentralize its operator pool. Phase 1 & 2 are delivered, they consist in an SoK for Decentralized identities and Verifiable Credentials, and an SoK and mechanism design for white-labeling resistance.
- Investigator in Ethereum Foundation grant *Concrete security of L2 zk-SNARKs*
- Contributor to the improvement of Cairo math libraries.

Education

2021 – 2022 Part III in Pure Mathematics

DPMMS, University of Cambridge

Master's thesis *Galois representations for modular forms.*

2018 – 2021 BSc in Pure Mathematics

Faculty of Natural Sciences, Imperial College London

- First Class Honours
- Programme Total 81.73%
- Top 5% of the cohort in second year ($\approx$ best 12 students out of 240).
- Appeared in the Dean's List ($\approx$ best 24 students out of 240) in the first two years.

Web Articles

April 2023 Algebraic Geometry Codes.

August 2022 A tour of Verifiable Secret Sharing schemes and Distributed Key Generation protocols.

with A.Garreta, M. Graczyk